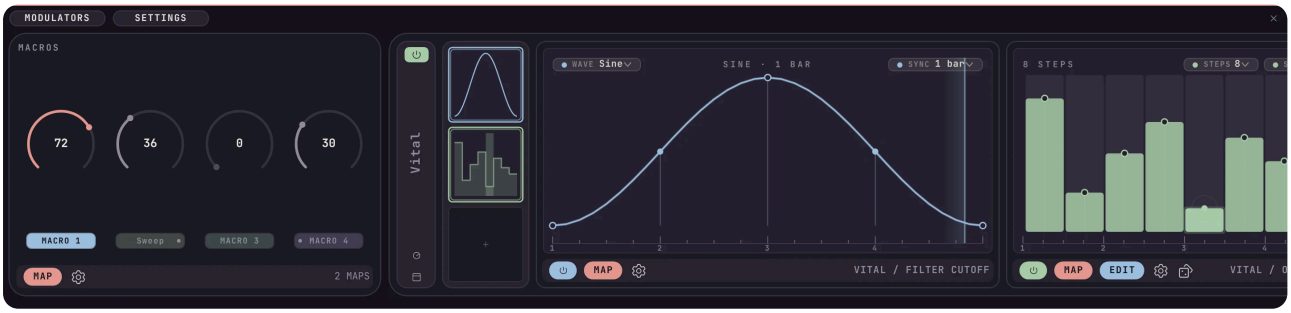
KERF AUDIO

USER MANUAL · VERSION 0.5.0 BETA

KERF MODULATORS

Seven modulators and a macro panel on every track, driving any plugin parameter on any track. Kerf Modulators docks into REAPER, drives the parameters your plugins already expose, stays locked to the beat, and saves with the project.

MACOS 11+ | WINDOWS 10 AND 11 | LINUX X86_64 | REAPER 6 OR LATER | FREE



Free and in public beta. Kerf Modulators has been tested and works on macOS, Windows and Linux, so far on small projects of around 10 tracks and 20 plugins. It has not yet been tested thoroughly in edge cases or larger projects, so expect bugs, and back up your projects.

DANNY WINCOTT, TRADING AS KERF AUDIO · GITHUB.COM/KERF-AUDIO/MODULATORS

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WHAT IT IS

REAPER can already modulate a plugin parameter, but only one at a time, through a Parameter Modulation window a couple of clicks deep, and with no single place to see what is modulating what. Kerf Modulators puts all of it on one docked panel.

Pick a track, see every plugin on it, drop a modulator into a slot, and map it to as many parameters as you want. Each track keeps its own modulators. Every plugin on the track gets a **block** holding up to **12 slots**, and any slot can drive up to **128 targets**, on its own track or on any other track in the project.

AT A GLANCE

Where it runs	Two of the eight hand off to REAPER's own audio-thread engine when their settings allow it. The rest run on the panel's clock with timing compensation.
What it touches	Only the plugin parameters you map, and by default only while the transport is rolling. Nothing is written into your audio.
Where it saves	Inside the `.RPP`. Send someone the project and the modulation travels with it, as long as they have Kerf Modulators installed.

Modulators

MSEG
A multi-stage envelope you draw yourself, looped or one-shot, free or locked to the grid.

LFO
Six shapes, free running or tempo synced, with jitter. Runs on REAPER's own engine where it can.

Macros
Four nameable knobs on every track, and the hub that every other modulator can drive.

XY Pad
Two outputs in one control, with a chaos wander that is free or locked to the beat.

Envelope Follower
Turns the level of any track into modulation. Ducking and sidechain on any parameter.

Steps
A step sequencer for parameters: up to 16 held values on a beat division, with randomise.

ADSR
A four-stage envelope triggered by the transport, a cycle or the start of each media item.

Waves
A curve library with real curve operations, your own saved shapes and undo.

Features

Native docking
A real REAPER docker: dock it, float it, move it between dockers. REAPER remembers where.

Many targets per slot
Up to 128 parameters per modulator, each with its own range and polarity.

Track-aware panel
Follows the selected track and shows one block per plugin, with horizontal scrolling.

Cross-track modulation
Drive plugins and macros on other tracks. Mappings survive reordering and renaming.

Chaining
Modulate a macro and everything it drives follows. Loops are impossible by design.

Native modulation
Sample-accurate LFO and follower on REAPER's own engine, chosen automatically per mapping.

Background running
Modulators keep running when the panel is hidden or another track is selected.

Themes
Four accents, including one taken from your REAPER theme, in light or dark.

Quick knobs
Two to eight hand knobs per plugin, with templates that every new instance picks up.

Timing compensation
Beat-locked modulators are evaluated ahead to cancel REAPER's render-ahead, so they land on the grid.

Project saving
Modulators, mappings, ranges and quick knobs are saved in the project.

Diagnostics
A built-in readout that shows whether a stutter is Kerf Modulators, REAPER or another plugin.

INSTALLING

Install with ReaPack, the free package manager for REAPER, from reapack.com. It works on every platform, keeps Kerf Modulators up to date, and on macOS it is the only way to install.

1. **Add the Kerf Audio repository**
   In REAPER, choose **Extensions › ReaPack › Import repositories...**, paste this address and click OK:

   ```
   https://raw.githubusercontent.com/Kerf-Audio/reapack/main/index.xml
   ```

2. **Install Kerf Modulators**
   Choose **Extensions › ReaPack › Browse packages**, search for **Kerf Modulators**, right-click it, choose **Install**, then **Apply**.

3. **Restart REAPER and open the panel**
   Choose **Extensions › Kerf Modulators: Show/hide panel**. It docks at the bottom to begin with. Drag it wherever suits you and REAPER remembers.

Installing by hand (Windows and Linux only)

Download `reaper_kerf_modulators-x64.dll` (Windows) or `reaper_kerf_modulators.so` (Linux) from github.com/Kerf-Audio/modulators/releases. In REAPER choose **Options › Show REAPER resource path in explorer/finder**, copy the file into `UserPlugins`, and restart REAPER. You won't get automatic updates this way.

Updating

ReaPack checks for new versions on its own. Apply the update and restart REAPER. Your saved curves and project data carry over untouched. To try fixes early, right-click Kerf Modulators in **Browse packages** and turn on pre-releases for it.

Removing it

In ReaPack, right-click Kerf Modulators, choose **Uninstall** and **Apply**, then restart REAPER. If you installed by hand, delete the file from `UserPlugins` instead. Your projects keep their modulation data inside the `.RPP`; it simply goes quiet. Saved curves are left alone, so delete their folder too if you want a clean sweep:

SYSTEM	SAVED CURVES
macOS	`~/Library/Kerf Modulators/wave-presets.json`
Windows	`%APPDATA%\Kerf Modulators\wave-presets.json`
Linux	`~/.config/Kerf Modulators/wave-presets.json`

> **Upgrading from an earlier build?** Delete any `reaper_kerf_modulators` file you copied in by hand, and on macOS any older `reaper_modular` file, before installing with ReaPack. REAPER loads every extension it finds, so two copies would both run.

ANATOMY OF THE PANEL

The panel follows whichever track you have selected. Pick a track with plugins on it and its blocks appear. Pick one without any and the panel empties itself, rather than leaving a stale track's modulators sitting there looking convincing.



THE TRACK'S MACROS ON THE LEFT, THEN ONE BLOCK PER PLUGIN, EACH WITH ITS SLOTS

The control bar

Every modulator carries the same row in the same order: power, Map, Edit, then the utility icons for curve type, settings and randomise. Learn it once and it reads the same on all eight.

CONTROL	WHAT IT DOES	RIGHT-CLICK
Power	Bypasses the slot and restores its targets	
Map	Choose a plugin parameter to drive	Copy, paste or delete this slot's mappings
Edit	Opens the full-size editor	
Settings	Range and removal for each target	
Randomise	Rolls new values, on Steps and XY Pad	
Block title	Click to collapse the plugin's block	Rename instance, delete plugin, clear all slots

> **Power is a true bypass.** It stops the modulator, blocks anything modulating it, and hands every parameter back to the value it held beforehand. It is not a pause.

WIRING AND ROUTING

A mapping is one modulator driving one parameter through a **range window**. The modulator always produces a value between 0 and 1; the window decides what that means once it reaches the parameter.

Making a mapping

1. **Click Map on the slot**
   The parameter browser opens, grouped by track and then by plugin, and paginated, so plugins that report thousands of parameters stay usable. The track you are working on sits at the top level; every other track is folded into **Other tracks**. Parameters you have already mapped are ticked.

2. **Choose a parameter**
   It joins the slot's target list. Repeat for as many as you like.

3. **Set the range under the cog**
   Drag the Lo and Hi handles. Putting Lo above Hi inverts the mapping, which is how you get ducking out of an Envelope Follower.

Working across tracks

A mapping remembers its target track by GUID rather than by position, so it survives reordering, renaming and reloading. Point a modulator on one track at a plugin on another and it carries on working. Find other tracks under **Other tracks** in the Map menu.

> **The Envelope Follower is the exception worth knowing about.** When its source is a different track, Kerf Modulators routes that track's audio into the target track's aux channels 3 and 4 with a send, because that is what REAPER's native follower listens to. The send is made for you, set to audio only, and the destination track is widened to four channels if needed. Remove the mapping and the send goes with it.

Chaining modulators together

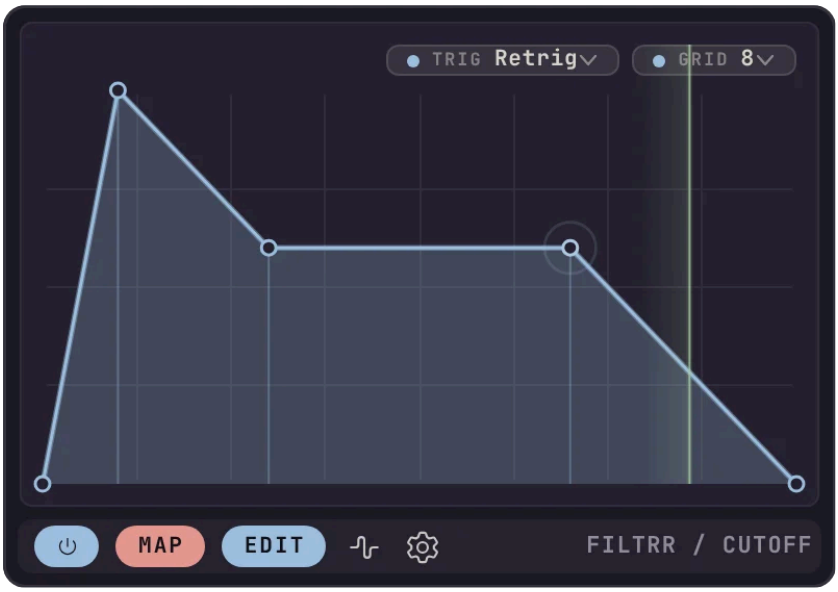
Map any modulator to a **Macro** knob instead of a plugin parameter, and everything that knob drives will follow. An LFO wobbling a macro that controls four filter cutoffs is one mapping doing the work of four. A macro cannot drive another macro, so there are no feedback loops to trip over.

THE MODULATORS

Eight in all: seven you add to a plugin's slots, and the Macros that every track has from the start.

01 MSEG

A multi-stage envelope that you draw yourself. Add as many nodes as you need, give each segment its own shape, and run it looped or as a one-shot, free or locked to the grid. One **Length** field on the editor face sets how long a cycle takes, as a value and a unit such as `2.0 bar` or `250 ms`. A fresh MSEG opens at `1.0 bar`.



CONTROL	WHAT IT DOES
Retrig	Restarts on play. A musical length locks to the grid; a real-time length (sec or ms) locks to the transport clock.
Free	Runs continuously and ignores the transport.
1-Shot	Plays once per transport start, then holds its final value.
Length	Nineteen units: `min`, `sec`, `ms`, `bar`, and straight, dotted and triplet notes from 1/2 to 1/32. The unit is the mode, so there is no separate sync switch. A bar follows the project's time signature at the modulator's position.
Delay	Offsets the start of the cycle, in beats.
Depth	Scales the output without moving any nodes.
Grid	Off, 4, 8, 16 or 32 divisions, which also turns on node snapping.
Loop section	Cycles part of the curve rather than all of it.
Jitter	Random wobble on value, timing, or both.

Double-click the canvas to add a node, drag to move it, double-click a node to remove it. The curve type button cycles the selected node's outgoing segment through Linear, Ease, Exponential and Hold. Drag a node onto its neighbour and the two snap to the same time, giving a genuinely vertical edge for a square wave or a hard reset.

02 Envelope Follower

Reads the level of a track and turns it into modulation. Point it at a kick to duck a pad, or at a vocal to open a filter as the singer pushes.



CONTROL	RANGE	WHAT IT DOES
Source	any track	Whose level to follow. Defaults to the modulator's own track.
Attack	ms	How quickly it rises to a transient.
Release	ms	How quickly it falls away afterwards.
Depth	0 to 100%	How much output it produces.
Gain	0.1 to 16×	Fine pre-gain, before smoothing.
Boost	0 to +48 dB	Coarse pre-gain, for quiet sources.

> **For ducking, invert the range.** Put Lo above Hi in the target's settings, so a loud source gives a low parameter value. That is sidechain behaviour on any parameter, not just volume. Leaving Gain at 1.0 and Boost at 0 keeps the follower on REAPER's own engine.

03 LFO

Six shapes, free running or tempo synced. This is the workhorse, and the one most likely to be running on REAPER's own engine.



SHAPE OR MODE	BEHAVIOUR
Sine, Triangle, Saw, Square	Reproduced exactly by REAPER's native engine.
Random	Sample and hold.
Perlin	Smooth drifting noise. REAPER has no equivalent, so it always runs on the panel's clock.
Rate, Phase, Jitter	Speed (free or synced), start phase, and roughness. Any jitter moves the LFO onto the panel's clock.
Retrigger	Resets the phase on loops and seeks, so a take sounds the same on every pass.
Free run	Keeps going regardless of the edit.

04 Steps

A step sequencer for parameters. Up to sixteen steps, each one a held value, advancing on a beat division you choose. Drag a bar to set its value, or roll the dice to randomise the lot; the highlighted bar is the step playing now. Randomising is the fastest way to a pattern you would never have drawn deliberately.



05 Macros

Four knobs you turn by hand: less a modulator than a control surface, and the hub every other modulator can be wired into. **Macros belong to the track, not to a plugin.** Every track has one macro module, first in the panel, there from the moment the track exists. Delete a plugin and the macros, and every mapping they hold, stay put.



- **Arm and map.** Click a knob to arm it, then Map. One knob can drive any number of parameters across any number of plugins.
- **Name your knobs.** Double-click a MACRO chip to rename it. A mapping then reads *Filter Sweep* rather than *Macro knob 3*, and one on another track reads *Drums › Filter Sweep*.
- **Modulated knobs** show a dashed arc. Switching a macro off freezes the knob and hands its parameters back, whatever is driving it.

- **Across tracks.** Each track lists its four macros under *Other tracks*. A dot on the left of a chip means another track drives this knob; a dot on the right means this knob drives another track.

06 ADSR

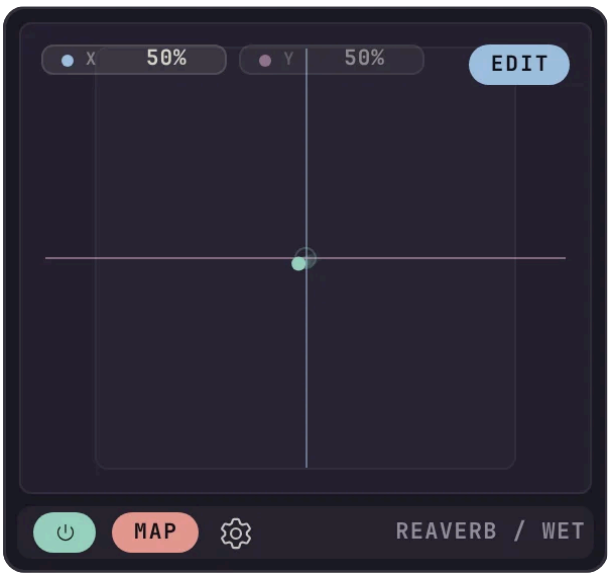
A familiar four-stage envelope with Attack, Decay, Sustain and Release, triggered by the transport rather than by notes. Play counts as note on; stop counts as note off.

ADSR SCREENSHOT TO COME

SETTING	RETRIGGERS ON
Cycle 0 (default)	Transport start, and every loop wrap, so it follows REAPER's loop.
Cycle above 0	Every N beats, locked to the transport.
Clip trigger	Every time the playhead crosses the start of a media item on the track. The arrangement itself becomes the trigger pattern.

07 XY Pad

Two modulators in one control. X and Y are separate outputs: arm an axis, hit Map, and that axis drives the parameter. Set a position, then let it wander. **Chaos** sets how far it strays and **Speed** how fast, in free mode. In **Beats** mode the pad glides to a fresh point every division, taken from the transport position, so it plays back identically every time.



08 Waves

A curve library that always retriggers. Pick a shape and set how long one pass takes with the same **Length** control as MSEG. A fresh Waves modulator opens at `1.0 bar`.



OPERATION	WHAT IT DOES TO THE CURVE
Inv	Flips the values top to bottom; timing is untouched.
Rev	Mirrors it in time, carrying each segment's shape with it.
Norm	Stretches it so the lowest point sits at 0 and the highest at 1.
Smooth, Sharp, Step	Forces every segment to Ease, Linear or Hold.
Quant	Snaps node times onto the grid.
×2	Squeezes the shape to play twice per cycle.
÷2	Thins out every other node. Repeat to reach a straight line.
Undo	Steps back through the last sixteen operations.
Save	Names the curve and adds it to your library.

The folder icon opens the gallery: factory shapes first, then your own. Right-click a saved curve to rename or delete it. Deleting one never breaks a project that uses it; the project keeps the curve and shows it as Custom. Saves go through a temporary file and a swap, so a crash halfway through cannot truncate the library.

06 · QUICK KNOBS

QUICK KNOBS

Knobs for the parameters you actually reach for, on each plugin's block: two, four, six or eight of them. No modulator is involved; you turn them and they write.

Right-click a knob to assign a parameter. Save the arrangement as a **template**, and every future instance of the same plugin picks it up on its own. Set up your knobs for a plugin once, and every copy you load afterwards arrives ready.

07 · NATIVE MODULATION

NATIVE MODULATION AND TIMING

Kerf Modulators has two ways of driving a parameter, and it chooses per mapping, continuously, as you work.

ROUTE	USED FOR	HOW IT BEHAVES
Handed to REAPER	The LFO on Sine, Triangle, Saw, Square or Random with no jitter; the Envelope Follower with Gain 1.0 and Boost 0.	Configured into REAPER's own per-parameter engine. Runs on the audio thread: sample accurate, immune to render-ahead, and fast synced rates such as 1/16 and 1/32 genuinely work.
Driven by the panel	MSEG, Steps, Macros, ADSR, XY Pad, Waves, and any LFO using Perlin or jitter.	Written from the panel. Beat-locked modulators are evaluated slightly ahead to cancel REAPER's anticipative render-ahead, so what you hear lands on the grid.

The switch is automatic and goes both ways. Add jitter to a synced LFO and that mapping moves onto the panel; take it away and it goes back. When a mapping is released, the parameter's previous modulation state is restored rather than wiped.

> **Kerf Modulators will not fight you.** Open a parameter's own Parameter Modulation window and Kerf Modulators stops writing to it while the window is open.

When you stop

By default modulators pause when the transport stops: parameters are handed back and everything rewinds, so each play starts from the same state. If you monitor live input through the FX chain and want modulation to keep running, turn this off in Settings.

> **Rendering to a file.** Only native mappings are part of an offline render. Everything driven by the panel keeps wall-clock time, so a full-speed render runs ahead of it and will not match playback. For an exact print of MSEG, Steps, Macros, ADSR, XY Pad, Waves or a jittered LFO, record the track's output in real time.

08 · THEMES

THEMES

Four accent colours (Coral, Teal, Blue, and one taken from your REAPER theme) and a light or dark switch, both in the Settings tab. They combine, so flipping between light and dark keeps your accent. A fresh install starts dark, and your choice is remembered per computer. Only the panel chrome changes; the editor screens stay dark, because the instruments shouldn't change just because the furniture did.

09 · SAVING

SAVING AND PROJECTS

WHAT	STORED IN	TRAVELS WITH THE PROJECT
Modulators, mappings and ranges	`.RPP`	Yes
Quick knob assignments	`.RPP`	Yes
Whether the panel is open	`.RPP`	Yes
Quick knob templates	REAPER settings	No, per computer
Saved curves	`wave-presets.json`	No, per computer
Accent and light or dark	REAPER settings	No, per computer

> **Undo does not cover modulator edits.** Kerf Modulators deliberately stays out of REAPER's undo system, because a partial undo state could silently wipe every modulator in a project. Ctrl-Z will not bring a deleted slot back.

DIAGNOSTICS

At the bottom of the Settings tab is a Diagnostics panel. You won't need it for normal work. It answers one question: when something stutters, is it Kerf Modulators, REAPER or another plugin? Start playback, let it run through the problem, then read the numbers.

READING	WHAT IT TELLS YOU
Tick rate	How often the modulation clock runs. Nominally 60 Hz. A much *higher* figure means the panel froze, then fired a burst of catch-up frames.
Tick cost	Our share of each frame, worst case in brackets. It should sit near zero.
Audible Δ/tick	**The one that matters.** How far the transport moved between frames. The largest figure is the longest freeze.
Proc Δ/tick	The same for REAPER's processing clock.
Audible / Processing	The two transport positions in beats; they differ by the render-ahead distance.
Delta / Comp	Measured render-ahead, and the compensation applied for it.
Param writes	Parameter changes since playback started.

Under about 110 ms looks perfectly smooth; around 340 ms reads as a visible freeze. The counters reset each time you press play.

> **A freeze is not proof that Kerf Modulators caused it.** Everything on screen shares one thread with every open plugin window, so one badly behaved plugin freezes the lot. If tick cost is near zero while the stall figure is large, the delay is elsewhere. Point the same modulator at a different plugin and compare.

TROUBLESHOOTING

WHAT YOU SEE	WHY, AND WHAT TO DO
The panel never appears	REAPER fails an extension load silently. Restart REAPER fully, then check there is exactly one Kerf Modulators file in `UserPlugins` and that it's the right one for your system.
The panel is empty	The selected track has no plugins. The panel hides rather than showing a stale track.
Modulation stops when I stop	That's the default. Turn off pause when stopped, in Settings.
A parameter won't move	Its own Parameter Modulation window is open. Close it.
A fast synced LFO sounds stepped	It has moved off the native engine. Set jitter to 0 and use a shape other than Perlin.
A cross-track follower is silent	The send needs four channels on the destination. Check the send survived.
Changes don't show after updating	An older copy is still in `UserPlugins`. There should be exactly one.
Timecode or meters stutter	Something is holding up the shared interface thread. Open Diagnostics and read the stall figure.

Reporting a bug

Report bugs at github.com/Kerf-Audio/modulators/issues. Questions and feature requests go in Discussions on the same page; new features wait until after version 1.0. Support is through GitHub only. Please include:

- Your system and version, and on a Mac whether it's Apple Silicon or Intel.
- Your REAPER version and the Kerf Modulators version from the Settings tab.
- Roughly how big the project is: tracks, plugins and modulators.
- What you did, what happened, and what you expected.
- A `.RPP` that reproduces it, if you can. It is worth ten paragraphs.
- For stuttering or freezes, a screenshot of Diagnostics straight after a run that showed it.

ABOUT THIS BETA

Kerf Modulators is free, and a thank-you to the REAPER community, which has inspired me since 2007.

It has been tested and works on macOS, Windows and Linux, so far on small projects of around 10 tracks and 20 plugins. It has not yet been tested thoroughly in edge cases or larger projects. That is where you come in: projects bigger than that are especially useful to test, and even a two-line bug report helps. Every report is read, and fixes come as time allows.

TESTING FIRST, FEATURES AFTER 1.0

This first release phase is for testing and bug fixing. No new features will be added until version 1.0 is out and stable. Feature requests are welcome in Discussions or Issues on GitHub, and they will be looked at after 1.0.

PLEASE BACK YOUR PROJECTS UP

Kerf Modulators writes into the `.RPP`, and a beta build can write state that a later build reads differently. Keep copies of anything you would be upset to lose.

Licence

Kerf Modulators is free to use, including for commercial work, under the Kerf Audio Freeware Licence. It is not open source. You may not sell or redistribute it; link to the official download instead. It is provided as is, without warranty. The full licence and third-party notices are in the repository at github.com/Kerf-Audio/modulators.

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KERF AUDIO

KERF MODULATORS · 0.5.0 BETA

Install with ReaPack, report bugs on GitHub, and back up your projects.

```
https://raw.githubusercontent.com/Kerf-Audio/reapack/main/index.xml
```

github.com/Kerf-Audio/modulators